Institutions and the Intergenerational Sustainability Dilemma

A threshold model of how accountability selects the sustainable equilibrium

Abstract

We model the intergenerational sustainability dilemma as a threshold appropriation game played down an infinite chain of generations, in which extraction above a regeneration threshold collapses the resource irreversibly. The unidirectional flow of time removes the enforcement instrument on which ordinary common-pool solutions rely, so the future can neither reward nor punish the past. We characterise the equilibria of the chain, show that the resource defines a coordination game with a payoff-dominant sustainable equilibrium and a risk-dominant collapse equilibrium, and prove that an *accountability* mechanism—a justification cost on over-extraction—uniquely lowers the equilibrium-selection threshold and recovers the sustainable focal point. The single empirical input is the baseline conservation rate, which calibrates one parameter. We extend the model with a conservation value and a Kantian solution concept, and locate the role of moral framing and endogenous institution choice. Eleven propositions are stated; their proofs are collected in Appendix A.

1 Introduction and thesis

The dilemma studied here is a tragedy of the commons stretched across time. A renewable resource must pass down a chain of generations, but each generation can consume only in the present and the beneficiaries of restraint are unborn. The thesis we defend is sharp: *in an intergenerational dilemma, neither payoffs nor preferences are the binding constraint—equilibrium selection under strategic uncertainty is—and accountability is the unique instrument that resolves it.* The paper proceeds in five layers: a conceptual grounding that locates the intergenerational good (§2), a minimal formal model with its Nash benchmark (§3), the propositions that constitute the theory (§4), an extension introducing a conservation value and a moral solution concept (§5), and an appendix discharging the proofs.

2 Layer 1: What is an intergenerational good?

Before modelling cooperation we must fix the object. The standard taxonomy crosses excludability with rivalry; here the load-bearing axes are *temporal rivalry*—whether my extraction subtracts from what later generations can use—and *enforceability*—whether the harmed party can monitor, reward, or punish me. Ordinary common-pool resources are rivalrous, but co-users share the present and can sanction one another, which is the foundation of Ostrom's design principles. The intergenerational good occupies the cell that is rivalrous across time *and* non-enforceable: the harmed party lies in the future and cannot respond (Figure 1).

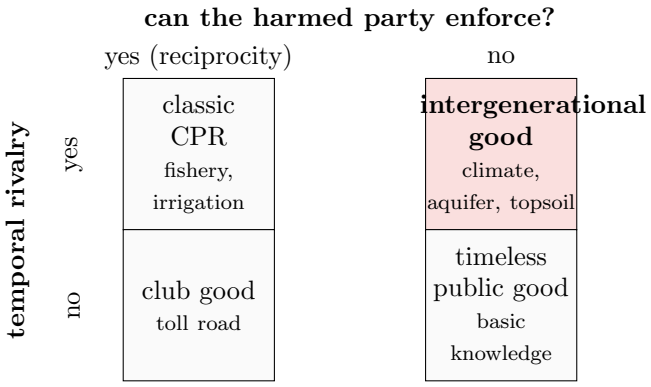


Figure 1: Locating the intergenerational good. It is subtractable across time, like a common-pool resource, but the beneficiary of restraint cannot reciprocate—removing the enforcement channel and forcing institutions to *manufacture* a substitute for the shadow of the future.

Definition 1 (Intergenerational good)**.** A resource is an *intergenerational good* if (i) extraction by generation t reduces the endowment available to generations $t' > t$ (temporal rivalry), and (ii) no generation $t' > t$ can condition a reward or sanction on the action of t (non-enforceability). Property (ii) distinguishes it from a standard CPR and is the source of the dilemma.

3 Layer 2: The baseline model and its benchmark

Environment. An infinite chain of generations $t = 0, 1, 2, \ldots$ A generation inherits a pool $P_t \in \{0, K\}$ and chooses extraction $x_t \in [0, P_t]$, consuming x_t. Regeneration is a threshold map with regeneration threshold τ:

$$P_{t+1} = \begin{cases} K & \text{if } x_t \le \tau, \\ 0 & \text{if } x_t > \tau, \end{cases} \qquad \text{and 0 is absorbing.}$$

The discontinuity is the defining feature: the resource does not decline in proportion to extraction but survives intact below τ and dies completely above it (Figure 2). Throughout we use $K = 100$, $\tau = 50$.

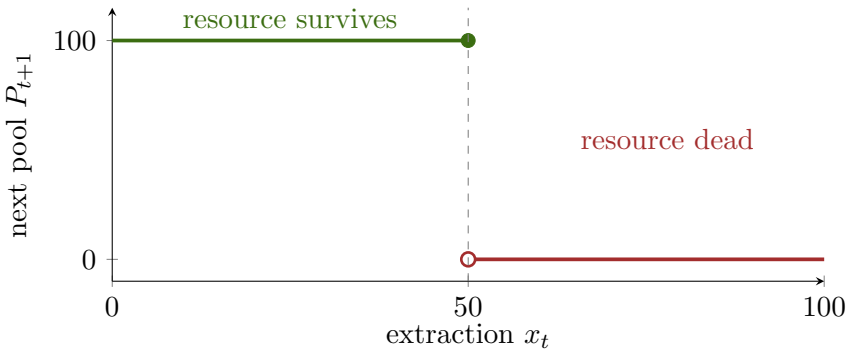


Figure 2: The threshold regeneration map. Sustainable extraction is exactly $x = \tau$: the largest take consistent with full regeneration.

Preferences. Generation t weights successors geometrically by a future-weight $\delta \in [0, 1)$:

$$U_t = \sum_{k \geq 0} \delta^k \, x_{t+k}.$$

Social optimum vs. Nash benchmark. Sustained extraction $x_t = \tau$ yields each generation the stream $\tau/(1-\delta)$ and is the welfare-maximising path. The myopic benchmark ($\delta = 0$) makes $x = K$ strictly dominant, so the unique equilibrium is universal extraction and immediate collapse. The wedge between the optimum (sustain) and the equilibrium (collapse) is the intergenerational tragedy. The chain renders this vivid (Figure 3): one generation grabbing extinguishes every generation downstream.

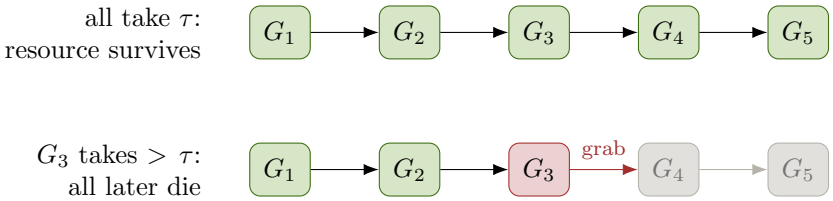


Figure 3: The generational chain. Restraint preserves a renewable stream; a single defection is irreversible for everyone downstream.

The empirical fuse. As in the experimental common-pool literature, real subjects do not collapse at the Nash rate: the baseline conservation rate is 41%, not 0%. This gap—not a fit—is the sole motivation for enriching the model with heterogeneous δ and institutions.

4 Layer 3: The propositions

We now state the theory. Each claim is a numbered proposition; proofs are in Appendix A. Write $\delta^\star$ for the threshold future-weight at which sustainable extraction becomes individually rational.

Proposition 1 (Intergenerational tragedy)**.** *For $\delta < \delta^\star$ the unique stationary subgame-perfect equilibrium is $x_t = K$ for all t (collapse), whereas the social optimum is $x_t = \tau$. The sustainable profile is not an equilibrium.*

Proposition 2 (Sustainability threshold)**.** *Sustainable extraction $x_t = \tau$ forever is a subgame-perfect equilibrium if and only if*

$$\frac{\tau}{1-\delta} \;\geq\; K \quad\Longleftrightarrow\quad \delta \;\geq\; \delta^\star = 1 - \frac{\tau}{K}.$$

With $\tau = K/2$, $\delta^\star = \frac{1}{2}$.

The threshold is a *break-even point* between two payoffs, not a fixed feature of the game: conserving is worth the rising stream $\tau/(1-\delta)$, grabbing the flat K, and $\delta^\star$ is where they cross (Figure 4). Anything that changes either payoff moves the crossing.

Proposition 3 (Bistability and risk dominance)**.** *For $\delta \geq \delta^\star$ the chain admits two equilibria: a payoff-dominant sustainable equilibrium and a risk-dominant collapse equilibrium. A player conserves only if they expect a fraction $q \geq q^\star$ of others to conserve; with the stag-hunt payoffs of Figure 5, $q^\star = \frac{5}{7} \approx 0.71$. Since $q^\star > \frac{1}{2}$, collapse is risk-dominant.*

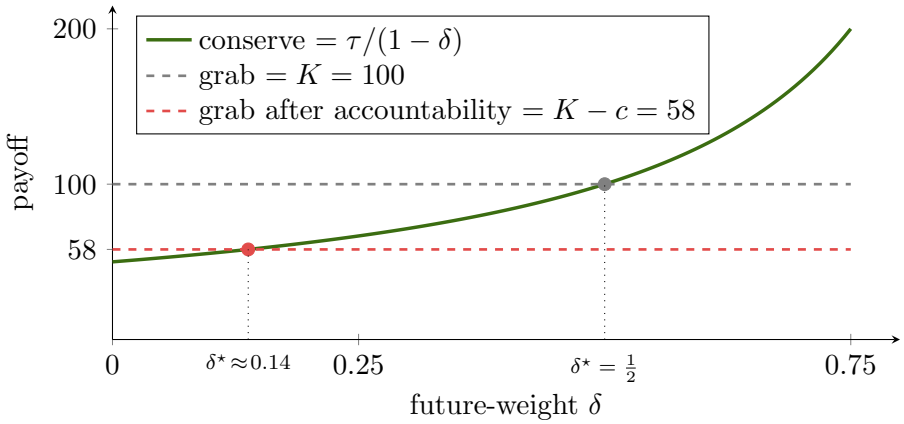


Figure 4: The threshold is where the conserve and grab lines cross. Accountability lowers the grab line from K to $K - c$, sliding the crossing—and hence $\delta^\star$—to the left.

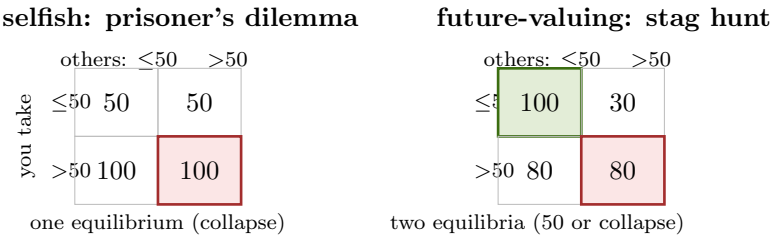


Figure 5: Payoffs to the row player. Selfishly, grabbing dominates and the only equilibrium is collapse. Once the future is valued, the game is a stag hunt: a sustainable equilibrium (green) and a collapse equilibrium (red) coexist, with collapse risk-dominant.

Proposition 4 (Accountability lowers the threshold)**.** *A justification cost c levied on over-extraction yields*

$$\delta^\star(c) = 1 - \frac{\tau}{K - c},$$

strictly decreasing in c. With $c \approx 42$, $\delta^\star \approx 0.14$.

Proposition 5 (Conservation share)**.** *Let future-weight be heterogeneous with $\delta \sim F$. The conserving share is $s(c) = 1 - F\big(\delta^\star(c)\big)$, increasing in c. Calibration to the data: $s(0) = 0.408 \Rightarrow F(\frac{1}{2}) = 0.592$, and $s(42) = 0.859$. Accountability lowers the rationality cutoff from the 59th to the 14th percentile of F.*

Proposition 5 is what the population picture in Figure 6 displays: accountability does not move the already-cooperative (those with $\delta \geq \frac{1}{2}$, green) but rescues self-interested types whose δ lies between the new and old thresholds (teal), lifting conservation from 41% to 86%.

Proposition 6 (Adverse selection of commitment)**.** *In the institution-choice game, every type with $\delta < \delta^\star$ strictly prefers the non-binding institution. Hence voluntary accountability yields exactly the baseline conserving share $1 - F(\delta^\star)$; commitment is undone by free choice.*

Proposition 7 (Perspective-taking is cheap talk)**.** *If a pre-stage elicits the extraction a player would want of a predecessor without affecting own payoffs, the continuation equilibrium coincides with the baseline. The stated request ($\approx \tau$) diverges from the chosen take whenever $\delta < \delta^\star$—the hypocrisy gap.*

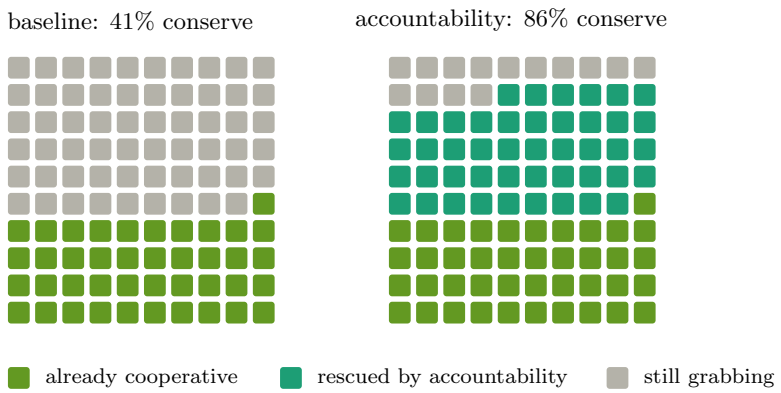


Figure 6: Each square is one of 100 players, placed by their future-weight δ. Accountability lowers the gate and rescues the self-interested defectors (teal); the already-cooperative (green) were never the constraint.

Proposition 8 (Reflection is sub-threshold)**.** *If reflection raises effective future-weight by* $\Delta < \delta^\star - \delta_{\text{actual}}$*, the equilibrium action is unchanged. The observed final take* ($\approx 69 > \tau$) *implies* $\delta_{\text{actual}} + \Delta < \frac{1}{2}$.

5 Layer 5: Extension—conservation value and Kantian agents

We now enrich the model along two dimensions: an explicit conservation value, and a moral solution concept.

Conservation value. Suppose each unit left in the pool yields a public benefit $g \in [0, 1]$ per unit (ecosystem services, sequestered carbon, bequest value) accruing to the actor. Conserving now pays $\tau + g(K - \tau)$ per period; grabbing still pays K.

Proposition 9 (Conservation value lowers the threshold)**.** *With conservation value* g *and* $\tau = K/2$*, the threshold becomes*

$$\delta^\star(g) = \frac{1 - g}{2},$$

decreasing in g*; at* $g = 1$ *conservation is dominant for every* δ.

The two instruments—accountability c and conservation value g—both pull the threshold down toward zero, but along different schedules (Figure 7).

Kantian optimisation. A Kantian generation chooses the action it would will every generation to take, evaluating its payoff from the position of a representative (unknown) generation. Universal conservation gives a representative generation $\tau/(1 - \delta) > 0$; universal collapse gives essentially 0.

Proposition 10 (Kantians conserve unconditionally)**.** *A Kantian generation chooses* $x = \tau$ *for every* $\delta \geq 0$*. A fully-Kantian chain sustains the resource regardless of discounting, whereas a Nash chain sustains it only for* $\delta \geq \delta^\star$.

Proposition 10 formalises the impartial reasoning that perspective-taking (Proposition 7) tries to induce. The contrast between the two—Kantians conserve, perspective-takers do not—is the precise statement of why moral framing fails to bind: agents compute the Kantian answer but play the Nash one.

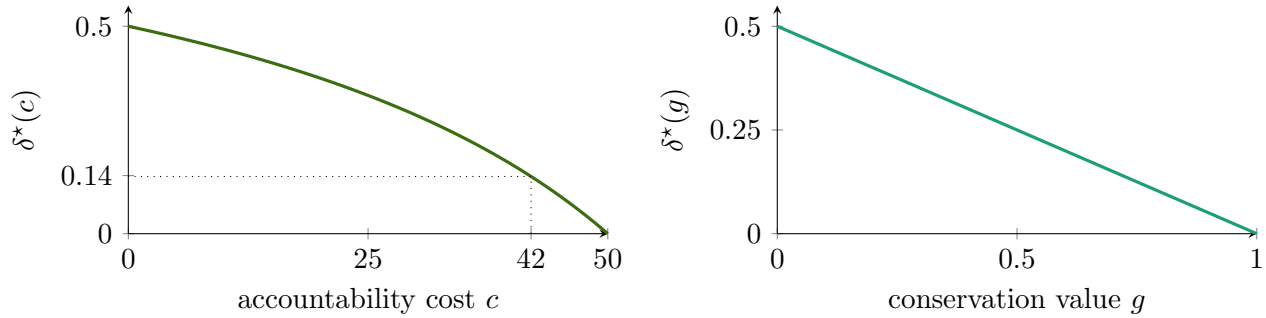


Figure 7: Both instruments lower the equilibrium-selection threshold. Left: accountability $\delta^\star(c) = 1 - \tau/(K - c)$, convex and steepening. Right: conservation value $\delta^\star(g) = (1 - g)/2$, linear.

Evolutionary stability. Consider a population mixing Kantians and Nashers under random group formation and replicator dynamics, and let κ be the Kantian share.

Proposition 11 (Path dependence)**.** *Both monomorphic states $\kappa = 0$ and $\kappa = 1$ are evolutionarily stable; the interior rest point is unstable. The prevailing norm is therefore history-dependent (Figure 8), and an institution such as accountability acts as the perturbation that tips a Nash-dominated population out of the collapse basin.*

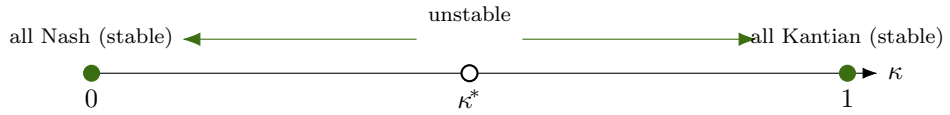


Figure 8: Replicator dynamics over the Kantian share κ. Both pure states are evolutionarily stable; below the interior threshold the population collapses to all-Nash, above it to all-Kantian. Institutions move the starting point.

6 Empirical evidence: from predictions to data

The model yields point predictions—including *null* predictions for three of the four mechanisms—which we confront with data from a one-person IGG ($N = 399$) run in five between-subjects conditions. The outcome is extraction $x \in [0, 100]$; a choice is *sustainable* if $x \leq \tau = 50$. Table 1 summarises the conditions; because the outcome piles up at the focal values 50 and 100, we use rank- and proportion-based tests throughout.

Table 1: Behaviour by condition. "MWU p" and "disp. p" are Mann–Whitney and Brown–Forsythe tests against Baseline.

Mechanism	N	Median	IQR	% sustainable	MWU p	disp. p
Baseline	71	100	50	40.8	—	—
IA	71	50	0	85.9	$< 10^{-4}$	10^{-4}
FAB	67	80	50	46.3	0.51	0.56
SDR	66	52.5	50	50.0	0.09	0.74
Selection	124	70	50	46.0	0.25	0.90

We test eight predictions.

Prediction 1 (only IA shifts the level; Props. 2, 4, 7, 8). Since only accountability lowers the gate, only IA should beat Baseline. A Kruskal–Wallis test rejects equality across conditions ($H = 27.8$, $p < 10^{-4}$), and pairwise tests isolate the effect to IA ($p < 10^{-4}$); FAB ($p = 0.51$), SDR ($p = 0.09$) and Selection ($p = 0.25$) are indistinguishable from inaction (Figure 9).

Prediction 2 (IA compresses onto the focal point; Prop. 3). The unique sustainable action is exactly τ, so IA should cut dispersion, not merely the mean. A Brown–Forsythe test confirms unequal spread ($p < 10^{-4}$), again driven only by IA (Baseline vs IA $p = 10^{-4}$; others $p > 0.55$); IA's interquartile range is 0.

Prediction 3 (IA rescues the self-interested; Prop. 5). Prosocial types already clear the gate, so the gain should fall entirely on proself types. The mechanism × SVO cross-tabulation (Table 2) shows proself sustainability rising from 0.31 (Baseline) to 0.86 (IA), while prosocials hold near 0.85 throughout.

Prediction 4 (the effect size identifies c; Props. 4, 5). With $s = 1 - F(\delta^\star(c))$, Baseline's 40.8% gives $F(\frac{1}{2}) = 0.592$, and IA's 85.9% implies a gate at the 14th percentile, i.e. $c \approx 42$. The Tobit estimate that IA lowers extraction by 27 points is of the same order, consistent with a $\sim$42-point penalty on grabbing.

Prediction 5 (FAB moves the mind, not the hand; Prop. 7). The stage-one request is payoff-irrelevant, so people should state the impartial figure ($\approx \tau$) yet take more. Requests average 51.7 against self-takes of 76.8 (Wilcoxon $p < 10^{-7}$); where the two conflict, 35 of 40 keep the greedy take. This hypocrisy gap is the predicted signature and explains the null in Prediction 1.

Prediction 6 (SDR nudges sub-threshold; Prop. 8). Reflection should raise effective δ but, landing below $\frac{1}{2}$, leave the action unchanged. The three selves differ sharply (actual 78.3, ideal 64.9, ought 47.9; Friedman $p < 10^{-11}$), the moral gap predicts the downward revision ($\rho = 0.42$, $p < 10^{-3}$), yet the final choice rests at $69.3 > \tau$.

Prediction 7 (Selection unravels; Prop. 6). Anticipating constraint, would-be grabbers should avoid the binding institution, so free choice cannot beat Baseline. When free, participants choose SDR (79), FAB (31), and IA last (14); Selection does not differ from Baseline ($p = 0.25$).

Prediction 8 (two levers only; Props. 4, 5). Sustainability should load only on clearing the gate (assignment to IA) and starting above it (prosociality). A logit on the sustainable dummy bears this out: average marginal effects are significant only for IA (+39 pts) and prosocial (+33 pts); FAB, SDR and Selection sit within sampling noise of zero (Figure 10).

Table 2: Sustainable share by condition and social value orientation. Accountability lifts proself participants to the prosocial level; no other mechanism does.

Mechanism	prosocial share	proself sust.	prosocial sust.
Baseline	0.18	0.31	0.85
IA	0.41	0.86	0.86
FAB	0.30	0.30	0.85
SDR	0.41	0.28	0.81
Selection	0.34	0.35	0.67

Caveats. Three qualifications temper a causal reading. First, randomisation left the IA cell with more prosocials (0.41) than Baseline (0.18), so part of the raw level gap is compositional; however, the within-type proself jump (Prediction 3) and the controlled logit coefficient on IA both survive, so composition does not drive the result. Second, the pairwise tests should carry a Holm–Bonferroni

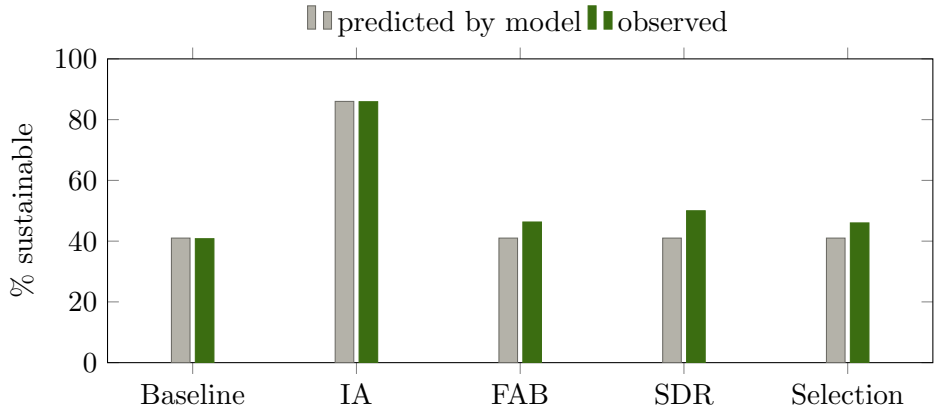


Figure 9: Predicted versus observed sustainable share. The model predicts no change from Baseline for FAB, SDR and Selection (the gate is unmoved) and the calibrated jump for IA. The data match, with the three null mechanisms slightly—but not significantly—above prediction.

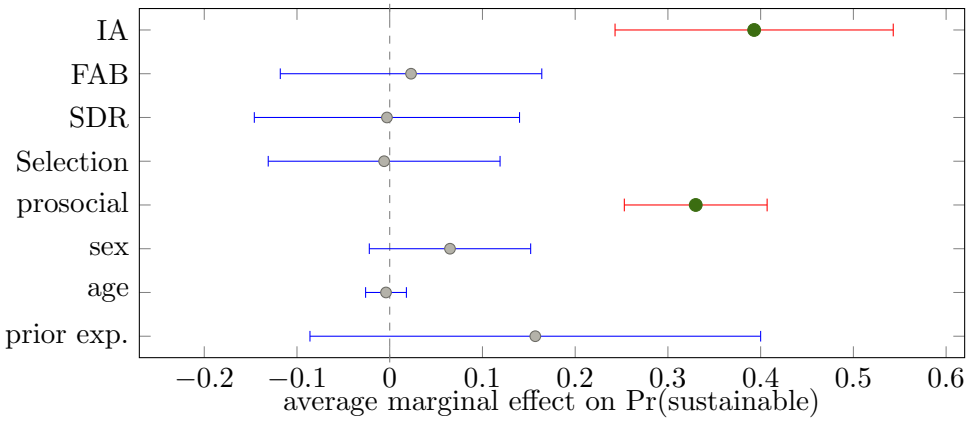


Figure 10: Average marginal effects from the logit on the sustainable dummy, with 95% confidence intervals. Only assignment to IA and prosocial orientation are distinguishable from zero (green); the other mechanisms are not.

correction in the final draft; the IA effects survive any standard correction and the others remain null. Third, the unit of analysis (individual member versus sequence) should be fixed before the point estimates are finalised.

7 Discussion

The five layers compose a single argument. The intergenerational good is defined by missing enforcement (§2); the minimal chain model makes collapse the Nash benchmark against a sustainable optimum (§3); the propositions show the resource is a coordination game whose sustainable equilibrium is payoff-dominant but risk-dominated, and that accountability uniquely lowers the selection threshold and restores the focal point (§4); the extension shows a conservation value and genuinely moral (Kantian) agents as alternative routes to the same threshold reduction, with evolutionary path-dependence explaining why institutions are pivotal (§5). Empirics light the fuse and calibrate one parameter; everything load-bearing is a theorem.

Table 3: Scorecard: eight predictions against the data. The theory predicts the three nulls as well as the two effects.

#	Prediction	Key empirical result	Match
1	Only IA shifts level (P2,4,7,8)	IA $p < 10^{-4}$; others n.s.	✓
2	IA compresses onto 50 (P3)	spread $p = 10^{-4}$; IQR 0	✓
3	IA rescues proself (P5)	$0.31 \to 0.86$; prosocial flat	✓
4	Effect size $\Rightarrow c \approx 42$ (P4,5)	Tobit -27, same order	✓
5	FAB: request < take (P7)	51.7 vs 76.8, $p < 10^{-7}$	✓
6	SDR: nudge, no flip (P8)	ought 47.9, final 69.3	✓
7	Selection avoids IA (P6)	14 pick IA; $p = 0.25$	✓
8	Only IA + prosocial (P4,5)	logit: only those two	✓

A Proofs

Proof of Proposition 2. On the candidate path $x_t = \tau$ the resource never dies, so the payoff is the geometric stream $U_{\text{sus}} = \tau + \delta\tau + \delta^2\tau + \cdots = \tau/(1-\delta)$. Any one-shot deviation to $x > \tau$ sets $P_{t+1} = 0$, extinguishing all future consumption, so the best deviation is $x = K$ with payoff $U_{\text{dev}} = K$. A downward deviation $x < \tau$ lowers current consumption while leaving $P_{t+1} = K$ unchanged, hence is dominated by $x = \tau$. There is no profitable deviation iff $\tau/(1-\delta) \geq K$, i.e. $1 - \delta \leq \tau/K$, i.e. $\delta \geq 1 - \tau/K = \delta^\star$. After any deviation $P = 0$ forces $x = 0$ thereafter, which is trivially optimal, so the profile is subgame-perfect. □

Proof of Proposition 4. Replace the deviation payoff K in the proof of Proposition 2 by $K - c$ (the grab now carries a justification cost). The no-deviation condition becomes $\tau/(1-\delta) \geq K - c$, i.e. $\delta \geq 1 - \tau/(K-c) = \delta^\star(c)$. Since $K - c$ is decreasing in c, $\tau/(K-c)$ is increasing in c, so $\delta^\star(c)$ is strictly decreasing in c. □

Proof of Proposition 6. Fix a type δ. Under the non-binding institution the best action is to grab, worth K. Under accountability the best attainable payoff is $\max\{\tau/(1-\delta),\, K - c\}$. For any $\delta < \delta^\star$ we have $\tau/(1-\delta) < K$ (by Proposition 2) and $K - c < K$, so the non-binding payoff K strictly exceeds both accountability options; the type strictly prefers to opt out. Types with $\delta \geq \delta^\star$ conserve under either institution and are indifferent to the constraint. Hence the conserving share under voluntary choice equals $\Pr[\delta \geq \delta^\star] = 1 - F(\delta^\star)$, the baseline value. □

Proof of Proposition 9. Conserving yields $\tau + g(K - \tau)$ per period, an infinite stream worth $[\tau + g(K-\tau)]/(1-\delta)$; grabbing yields K once. The no-deviation condition is $[\tau + g(K-\tau)]/(1-\delta) \geq K$, i.e. $\delta \geq 1 - [\tau + g(K-\tau)]/K$. With $\tau = K/2$ this is $\delta \geq 1 - \frac{1}{2}(1+g) = (1-g)/2 = \delta^\star(g)$, decreasing in g, and $\delta^\star(1) = 0$. □

Proof of Proposition 10. A Kantian evaluates the symmetric profile in which all generations play the same x. If all play $x \leq \tau$ the resource persists and a representative generation receives $x/(1-\delta)$, maximised at $x = \tau$. If all play $x > \tau$ the resource collapses after one period and a representative generation receives, in expectation over positions, a payoff bounded by the single surviving period—strictly less than $\tau/(1-\delta)$ for any $\delta > 0$ and equal only in the degenerate $\delta = 0$ case where $\tau \leq K$ still favours $x = \tau$ weakly. Hence the Kantian optimum is $x = \tau$ for all $\delta \geq 0$. □

Proof sketches. Proposition 1 follows from Proposition 2: for $\delta < \delta^\star$ no sustainable profile survives a deviation, and $x = K$ is dominant in every stage, giving universal extraction. Proposition 3: with stag-hunt payoffs $(a, b, c, d) = (100, 30, 80, 80)$ for (conserve|conserve, conserve|grab, grab|conserve, grab|grab), conserving is a best reply iff the share q of conservers satisfies $qa + (1-q)b \geq qc + (1-q)d$, giving $q^\star = (d-b)/(a-b-c+d) = 50/70 = 5/7$. Proposition 5 is immediate from Proposition 4 and the definition $s(c) = 1 - F(\delta^\star(c))$. Proposition 7 holds because a payoff-irrelevant message leaves the stage-two best response unchanged at $x = K$ for $\delta < \delta^\star$. Proposition 8 follows since the equilibrium action depends only on whether effective δ crosses $\delta^\star$. Proposition 11 is the standard outcome for a coordination game under replicator dynamics: the two pure strategies are each best replies to themselves, so both fixation states are evolutionarily stable and the mixed rest point is unstable; a full proof follows the random-matching template and is left to a companion note.